

ActInf Textbook Group ~ Cohort 9 ~ Session 3

(Chapter 2 and 3) ~ 9/4/2025

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Cool. >> Hi everyone. Uh, welcome. Thanks for joining us the textbook reading group. And today we were going to look at chapter 2 in the first half of the textbook. Um, and I I I gave this kind of introduction of the book uh a couple weeks ago for chapter 1 uh with with another sort of group of folks. But uh a brief recap is that the first half of the book is essentially providing like uh sort of theoretical foundations for active inference and kind of covering a lot of the various uh associated fields and theories. So we introduced to the basian brain hypothesis. We're um told about predictive coding and message passing in chapter 5 and all of these different things. Marov blankets come up later. Um and then it's in the second half of the book that's more about direct modeling. And so with what Daniel mentioned earlier, I actually uh I'm curious about what he it sounds like he just made this. So probably a learning resource that would be useful for anyone who's interested in getting into the coding themselves. Uh but yeah, so for for chapter 2, uh following from chapter 1, this is where we start taking the uh low road to active inference. And this is uh starting with sort of a basian framework. Uh and so for those unfamiliar, um Baz's rule is essentially a rule that describes how we can find we can kind of update uh our beliefs uh our prior beliefs about a given phenomena given that we have some kind of joint distribution uh which is a generative model that can be used to predict phenomena. So um whenever we take a joint distribution for example the the probability of X comma Y . So uh two random variables x and y uh we we we can kind of uh decompose that into uh this this nice really sort of concise theorem uh that is known as basis theorem that if we are able to update particular components of that then it allows us the appropriate rules for updating the other components such as uh if you have a x representing beliefs about hidden states of the world and you have y which is observations received from that world then based on new observations you can then update your beliefs um so given a new y you can update your beliefs about X and the relationship between X and Y that so that's that's the that's sort of the general framework it's it's used very broadly it's existed for for decades it's used uh

very commonly in all kinds of fields, machine learning like financial modeling uh very much used in neuroscience as well. Uh not just in an active inference framework. So so again it's just kind of speaks to the interdisciplinarity uh of active inference is sort of the depth of how far it's traced itself to roots in various other fields. Um and so the the the second chapter largely sticks with that. we we get introduced to other concepts such as um surprise and basian surprise which are were which are unfortunately and slightly confusingly at first glance like they're two distinct concepts. Um I I find them interesting to think about in that uh basian surprise would be taking uh the what's called the coldback leler divergence uh between uh posterior distribution and a prior distribution. So it's kind of like saying okay we received a new observation we update our new beliefs our posterior beliefs. uh how different are our new beliefs given that observation versus uh our prior beliefs that we had before we received that observation. Uh if they're very different that means a lot of belief updating needed to happen. Uh right. So what could happen is that uh they they provide this as an example in the textbook. Um but if you were to strongly commit to a particular belief, if the waiting was such that uh you know you were you were fully convinced that uh that you were not hungry, you are sad in a in like a categorical probability distribution of you know am I hungry? Am I sad? Um like if you were fully convinced that you were sated uh then that means that if you received an observation such as oh your stomach is growling which would you know we could speak about that like we might think intuitively oh that would mean you're you must be hungry right but if someone were really fully convinced with like a pro full probability of one that they were sated regardless of any new incoming observations then your outcome your posterior beliefs will still be that you are sad despite receiving that observation. So, it's kind of like e even though um even though you might not expect to receive this observation that that you you know your stomach is growling, you're still firmly convinced that that you're sated and therefore no belief updating happens. Um I so I I I recommend looking at that. I think it's in box 2.2. Um but but this idea of basian surprise of how much belief updating happens versus surprisal itself which is a bit simpler. It's just the marginal probability of a new observation. So like given a generative model how well does that model predict that your stomach's going to growl you know and that's it right? So that just that so it has nothing to do with your beliefs about uh if you're hungry or if you're sad. It has nothing to do with X. It's only the Y. So it's a which is why we call it a marginal probability. Um you know marginal meaning we kind of can decompose the the model into just these these marginal parts. So um so surprisal is really just the negative log probability of an observation that you receive. So

you could receive so so it could be very surprising to you that your stomach is growling and yet you are highly convinced that you're you're sated nonetheless. Right. Um I like this example because uh for those who are interested in the field of uh computational psychiatry which there are many who who come into this field um come into active inference because they're interested in or studying computational psychiatry and are looking to kind of add more tools to their toolkit or another framework which I I think that active inference is very it's it's very wide and I think it brings a lot to computational psychiatry. You can imagine that someone who does not uh in everyday terms does not listen to the new observations that are coming in. You know, their stomach is growling and yet they believe they're sated. It it's not terribly far away from very common, you know, definitions of what what it means to to uh have delusional thinking or an inability to update one's beliefs despite evidence to the contrary, which uh is is kind of widespread uh in a lot of uh sort of psychopathologic like symptomatology, right? So um you know someone with with schizophrenia or someone who is hallucinating who's fully convinced that something is there uh whenever it's not right or someone might tell them that it's not but that you know thus providing them with contradictory evidence to their beliefs and yet they still hold. So um it it's more difficult in the case of hallucination because in the case of hallucination you are in fact receiving an observation that something is there right. So it's just a it's a reminder that this is not simply a uh sort of prefrontal cortex uh only sort of oh I just have to change the way I think and I'll be fine. this this refers to kind of deeper like aberrances in in the brain and in uh in sort of um neuronal message passing, right? And so things that can happen that can end up producing a hallucination. So um that that's that's why I kind of point to like okay well you you can receive very surprising observations and yet there's no basian surprise in the sense that you could still not update your beliefs, right? So there's belief updating and there's what the model actually predicts. U those are sort of the core some of the core aspects of uh kind of understanding the low road to active inference and and the the basian framework and then it's in uh especially later chapters chapter 4 uh where we're introduced to expected free energy and different breakdowns of free energy. um chapter seven which starts talking about matrices and and sort of these um categorical probability distributions. So actually taking Baz's rule and actually putting it into uh a model that goes beyond just having a likelihood. So so chapter two is for anyone who's new to a basian brain uh a basian framework it it'll be a significant amount to digest. uh for those who are already familiar, they're not going to receive anything too dramatically new until they start getting into later chapters and looking at how the basian

framework actually comes into more full play um and active inference. I noticed that someone uh someone's hand was up over time and so u >> yeah is is that okay if I I jump in? I I think these you have great explanations here and I really appreciate that. um trying to get an understanding of how we get from from bay to active inference. But but the my question is is you were talking about the kullback divergence. I was thinking of it as being the the distance between your between the Q which is like the approximate posterior and the actual posterior. But if you're saying it's that it's the difference between your prior and the posterior, I can wrap my head around that a lot better. It's just sort of the cost of your belief updating. Is it? But but I I maybe I'm confused which is which is it is it the prior to the posterior or is it the the approximate posterior to the true posterior? >> Yeah. Yeah. No, I I see what you're saying and it depends on like kind of the the modeling situation at hand. uh is how I would say that. So the the formal uh kind of definition the way it's described here is the posterior uh and prior uh like that's what's given in chapter 2. But then that said like um in in whenever you're actually implementing this within a model, it's there that it becomes significant to look at the the other uh perspective because in chapter 2, we're not actually yet getting into variational methods, right? So we're not looking at an approximate posterior. So what you can do is that you can in in very simple examples like the the frog and the apple example that we're given for example you can it's very tractable to just directly compute surprise and bayesian surprise right like we you could do it by hand. Uh so it's in those cases like there is no variational um posterior even computed in the first place. You're just given it like an exact posterior, right? So it's it's whenever you start getting into using the variational posterior that that this kind of additional dimension of looking at things and then actually if you could if it were the case that like say your model itself contains like what it believes to be the the true uh you know set of beliefs and then meanwhile you're using this variational method to update a variational sort of posterior then you would do you could use KL divergence to try and compare like but but that would still be relative to that model like does that make sense? Um yeah. >> Yeah, I think so. Thank you so much for clarifying that. >> Yeah. >> And when we get to the variational part then I'll I'll probably have more questions but for now this is this is great. Thanks so much. >> Yeah. No, absolutely. And I I totally welcome questions and I think that usually they end up providing a lot more information to to other people who might be considering the same thing. Um and and yeah, it's that's that's sort of a big move to start using variational um methods in the first place. You're just like kind of in a qualitatively different paradigm. Even though doing exact bayesian inference versus variational bayesian inference, it's like oh they share

the basian inference part, right? But but uh but yeah, once you start using variational, it's like there there's a couple additional assumptions that get made along the way whenever you're using variational methods. Uh for anyone who's into computational modeling and kind of like really breaking that down thoroughly, that does mean that you know unfortunately you do have to rely upon a couple of extra uh assumptions in order to view the outputs as is valid. uh one way of of doing variational basian inference is to use this kind of active inference framework and the particular ways that they do that. So you'll see in later chap I don't think they get into it in chapter 2 at all but um later they'll start talking about how free energy itself the way that sort of works is a is a variational method is that um using uh what's called Jensen's inequality. So I I'll kind of speed it up because it there will be more to get into whenever you read about it in later chapters. Uh but you it's kind of like by using particular mathematical I hesitate to call them tricks because they're not tricks like they're like proofs but still it's like by being able to exploit particular um aspects of the mathematics you're actually able to create an upper bound on surprisal. And so even though you can't directly compute surprisal, you can at least compute an upper bound on it. Uh meaning like well if you can if you can continue reducing that upper bound then you're getting closer and closer to actually minimizing your surprisal as well. Uh for sure and so that upper bound is itself free energy and it is tractable. Right? So it's that's the entire intention of using variational methods is to just render a lot of these computations tractable and possible and perhaps even more efficient um than than in cases where you just wouldn't be able to because you're trying to do exact inference. Um it's also very relevant for those who who are concerned with like biological plausibility of these models. um it it fits really well with different kinds of research in that area because of course like the notion that the brain computes exact basian inference whenever we know that many of these problems are not tractable just doesn't hold up very well in like kind of a realistic uh sort of setting. And so part of the the biological plausibility is is this uh aspect of the theory or framework of active inference that uh very we we there must be something variational going on uh so to speak, right? Being able to kind of use mean field approximation and marginalizing out different components uh of a model and updating them, you know, potentially separately or using these intricate message passing schemes. So the goal is to sort of render intractable problems tractable while simultaneously cohering with research that has been done uh on the brain and how message passing seems to occur, how neurons seem to communicate with one another. Um most research even outside of active inference tends to view this as like a kind of hierarchical message passing process. there multiple systems that

are kind of passes passing messages. So we have these kind of uh descend and similarly the there will be more kind of mechanical aspects of this. Um, I think especially in chapters four and five, uh, and definitely five for anyone who's interested in actually like aligning this with with, um, like deep, uh, pyramidal cells and and and like the kind of the more neurobiological aspects. Um, feel free to skip ahead to to chapter 5 if you feel like you have a a comfort comfortable grasp of everything prior to that chapter. Um but but yeah, so so being able to kind of have these these things cohhere with the brain while simultaneously keeping attractable um yeah is the gist. So um anything else I would kind of highlight about chapter 2 because it it's there there's quite a bit going on and actually I realize now we do get introduced to um we do get introduced to expected free energy already in chapter 2. um and we're introduced to variational free energy. So um I'll I'll just make a couple comments that that both of them are variational methods. Um so we do have variational free energy and expected free energy. They are in fact two different ways of computing free energy. uh variational free energy would be sort of relative in real time to the model given where it is at uh versus expected free energy which hinted at by the word expected is sort of a free energy of the future so to speak. Expected free energy is what's used whenever an agent uh does what's called com uh computing policies. So anyone who's uh familiar with reinforcement learning that will become or uh that'll be very apparent. Um like whatever your intuition says it's correct. Uh policies or you can think of policies as actions or more precisely sequences of actions. Uh right? So so if you if you think of like a discrete case like a a mouse in it in a team and it can choose to go left or right. Um and then there's also like another space it could go to. So a policy might be, you know, move to this part of the maze, then move to the left arm of the maze over the course of like two time steps like sort of into the future, right? And so that that doesn't involve um only computing variational free energy in the moment, but it actually also includes uh predicting what will I expect in the future. And that's where expected free energy comes in and why it's relevant to have this second um equation that is it's highly related. They're both variational. They're both using the same components of the model which you can liken to everyday concept. A likelihood model is very similar to what allows us to have perception. Um and then uh having a state transition a state transition matrix is very comparable to the idea of just having a temporal dimension having an understanding of like well this le led to this so then I can predict that this will happen next if I do this like having that kind of temporal dimension of of how you relate things. Um so yeah with with expected free energy um also important there is that uh we can interpret expected free energy a little bit

differently than in variational in that uh whenever we look at it's equation 2.6 and I apologize usually I share my screen but I've been having some issues with that today. Um but equation 2.6 six, which is on page 33. Um, and it's it's a bit of a hefty equation there, but it's really just being broken down in uh multiple ways. And I think especially that first line where we're looking at um information gain and pragmatic value. Uh, interestingly, as opposed to variational free energy where we really just think of this as well, we're trying to reduce complexity while improving accuracy. Um, here with expected free energy, we're trying to uh maximize two terms which each of which are scored negatively. They're both multiplied by negative one. So technically by maximizing these respective terms we're uh minimizing or reducing the overall quantity whenever we put them together uh and multiply them by each by negative one and then and then combine them. Um so information gain uh and pragmatic value pragmatic value is significant in that there's this um this component C that we can see there um the C is is a representation a probability distribution uh of the observations that the model expects um just as a prior. So it's like a prior over observations. So the way that this is formalized in active inference is to say that um priors over observations they they tend to get likened to the idea of preferences and not necessarily preferences in an arbitrary sense like oh I prefer this kind of dessert versus this kind. Um preferences also strongly relate to the notion of something like homeostatic set points for example. So, so if you had a kind of continuous space like predictive coding model um that had to do with an agent's beliefs about its um you know blood oxygenation levels, right? Or its heart rate. Um having preferences over sort of like well-regulated levels for those different kinds of physiological measures would be really crucial for the agent's survival, right? Um, so being or even just bringing up the the hunger like the the the kind of slightly silly but still very real um example I brought up earlier of like your stomach growling or not like we tend to prefer our stomachs aren't growling. I again I don't want to put it in such an arbitrary sense because there's no assumption that like you are like fully um consciously rationally thinking of things in this way. It can be that uh different parts of the body, different parts of the brain depending on how you want to sort of divide up the marov blanket as it were um each one of them having their own kind of processes and sort of inner model involved and have their own sort of so to speak preferences or priors over observations that they would need to uh achieve through action in order to maintain the overall organism's survival. Right? So, so the for those interested in like you know sort of um chaos theory or dynamics or just looking at like non-equilibrium steady states this is very relatable to the idea of an attractor right so so whenever you take these these sort of attractors or you know

the the the brain the body is attracted to you know having the body at roughly 98.1 degrees uh Fahrenheit uh you know is is a is sort of normal normal or healthy um um you know body temperature, right? Um that figures into expected free energy and saying like well expected free energy ends up being scored over each of the policies available to the agent. That's what that that G parenthesis π represents. π is is your overall like your policies your policy space. Each potential policy gets scored by G . And then we see that pragmatic value, that difference between the actual observations you expect given that policy versus your overall preferences. Basically, do I think taking this action is going to get me what I prefer? Right? Is is this action going to help me achieve a regulated heart rate or a sated uh you know hunger level or stomach or or or you know level of glut glutamate uh or or otherwise um or you can think of it more whimsically like if I go to this store will I be able to you know get that dessert that I really like uh right so that that's what kind of pragmatic value is we want to maximize it meaning like if I take this action will I be able to expect that I will be able to kind of get what I want or realize my preferences or realize my priors over observations that's why we have that that that conditional um the the just if if you ignore the the expectation in the log it's just a probability of a sequence of the sequence of observations you expect if you take this action conditioned on uh and thus related to uh your preferences. So essentially pragmatic value that that's uh you know in reinforcement learning you can almost think of that as like that's sort of your reward function right this is how the agent is able to sort of plan and and figure out you know if I ex you know if I take an action what observations will I be able to expect and then I can compare them to what I actually want and that's kind of how you score that uh meanwhile however we also have information gain and this is part of what attracts me uh like myself and the the kind of work that that I do because some of what I do relates to uh reinforcement learning uh in addition to computational psychiatry um I've noticed in reinforcement learning it's sometimes very difficult to kind of handle the the so-called explore exploit dilemma uh which is where uh you know whenever you have a model whether you're thinking of it as a human being who you know might be suffering from from some kind of um mental um you know disorder or or trouble or it could simply be you know creating some kind of robot that navigates a grid space or something. Um it's difficult to say what should define exploration and and for those who are interested in philosophy there's always this question of like well why don't people just kind of you know get the get as many resources as they need and then just stay in their room you know that that minimizes all potential surprisal as much as possible right shouldn't shouldn't that be what we do um of course I would say like well if that doesn't

match your preferences then no you won't do that uh and we don't tend to prefer for uh staying in dark rooms 24/7, but it potentially for biological reasons, but that's that's its own uh conversation, I guess. U but all that is to say that um with with active inference, you you get this like kind of nice formalized notion of what exploration is uh and and it's already included with pragmatic values. So where whereas in reinforcement learning what'll happen is people will focus on what active inference calls pragmatic value and then maybe you'll have some kind of rule in there that says like oh the agent should do something random 10% of the time just to make sure it doesn't get stuck in a local uh uh optimum right like the idea of like you know suboptimality is sort of relative to to a model uh as well as to to the observer, right? So so someone who um you know in the case of schizophrenia who is hallucinating, right? Like if their brain is kind of computing that hallucination as it were given that their their their brain's makeup at that time sort of the the the the kind of figurative generative model that it's working with to produce that hallucination in a sense is kind of optimal to that model despite from an external observer's perspective say a psychiatrist or or a therapist or or just another person would say like oh that doesn't seem correct relative to the environment you're in. Right? So um all all that is to say that um you know we have pragmatic we we we there there's the optimality that one can kind of get stuck within right um and so it's it's theorized that it's through exploration that that people are able to sort of get out of local optima or find increasingly hopefully ideally increasingly uh more optimal solutions to to everyday problems. And that's sort of a foundational aspect of adaptive learning. So, um, so by being able to actually have information gain as a piece of this, uh, is kind of like saying like this is how it's saying that this is how the model works, but it also says we're not just doing something random. We're doing something that actually continues to further the goal of reducing surprise, of minimizing free energy. Um it it so so it starts to to kind of make more sense rather than just viewing it as some kind of random number generator process which is like quite explicitly how that would be be done in reinforce reinforcement learning models on average. So with that that information gain term, it's kind of um I mean it's essent essentially just taking the expectation of the the KL divergence between uh you know here's what I so it's it's a Q of X conditioned on Y and π versus Q of X conditioned on π . So we already see like the only difference between those two things is that y is actually incorporated into that uh the the first distribution of those two. So, it's kind of like saying, well, here's what I thought uh was going on or what I what I think about the hit here are my beliefs given uh here here's how I believe my the hidden states of the world relate to what I do, my policies. Here's what I think I can do that changes

the world. However, here's now what I think how I think that works given the new observation. I'm apologize. I kind of got myself in a tongue twister there, but um it's just to say like given this new observation, what do I think about the relationship between hidden states of the world and what I can do? So that means that by trying new things, you can try and elicit new observations that can then help you to learn more about the world, right? Which is interesting because it kind of conceptually it's almost like an inversion of uh variational free energy, right? with variational free energy, it's like, well, we should be self-evidencing, so we should just be trying to do essentially the same things. But but in that case, we we never there's no guarantee that we'll ever actually like um accomplish our our goals uh or come closer to accomplishing any kind of goals or preferences, whether that be homeostasis itself or it be, you know, a human being's like long-term career trajectory or some such, right? So, um that's sort of um that's sort of the the the gist, uh the overview of expected free energy and and not just like how to sort of interpret those first two terms, information gain versus pragmatic value, but also kind of understanding like this is a very um kind of self-consistent process that occurs that doesn't rely upon sort of random processes, right? like you you can try it you can include like an error term uh in different places with predictive coding models like this is the unexplained variance but um but there's no forced like oh 10% of the time do something random and that's sort of the what some would argue would be the sort of elegance of of planning in an active inference scheme as opposed to you know comparison to some others Mark yeah you had your hand up again if you want to go for Sorry, I was on mute there for a second. Yeah, thanks. Uh, this is just another great explanation. I just want to make sure I understand this. So, when we talk about expected information gain, right, we're we're talking about the the distance between again sort of like prior beliefs about policies uh and and the posterior beliefs about policy. So like as the farther these are apart uh the greater information was gained through but through the through the update and that is then inversely proportioned to the free energy. Is that right? So there's a negative sign. So the so the the more I learn I guess the the bigger the update the smaller the free energy. Is that right? So I I've resolved uncertainty. Is that a fair way to look at it? >> Okay. So >> Okay. Okay. So I was thinking of it in terms of like there's a there's a cost there there's entropy associated with that belief update though. Is is that right? Am I maybe I'm conflating. >> No, you're you're >> because I'm almost there, I think. >> Yeah. No, you're totally fine. And in fact, I mean, you know, again, looking at the entire equation, it's like there are other ways of kind of sort of breaking this up. And then you see like with expected ambiguity immediately below that h like that is where we

are explicitly like looking at this in terms of like kind of entropy functions. Um but but if we just constrain it to looking at it as information gain first off yes you're absolutely right the I the idea is that if you if you sort of maximize or increase information gain if you learn more about the relationship sort of between the world and observations and and policies given that there's a negative sign there that does mean that the more we increase information gain the more expected free energy will decrease. um to put to briefly put that into the perspective of some of what I was saying earlier um if free energy is an upper bound on surprise right then we're going to want to minimize uh free energy so in this case by learning more you would indeed be reducing free energy and then from reducing surprisal right so it's I mean it's um we also remember like you could improve information gain like immensely yet simultaneously have like virtually no pragmatic value, right? Like um oh I almost want to say like trigger warning or something, but I realized I came up with like a very dark example, but it's like well someone doesn't know what it's like to you know I'll I'll lighten it up a little bit but it's still a bit dark. Uh, someone doesn't know what it's like to be unconscious. So, they're like, "Well, I could knock myself out, right?" Like, is that you could learn a lot? Maybe, maybe not. You realistically, no. But, but someone could have that belief if you just kind of localize it to this is their thought space right now. Uh, this is sort of their generative model and the problem they're trying to figure out. Like, is it truly worth it to do that in order to learn that, right? Um so so if it's if the answer is no or or at least if that the way that policy of knocking yourself out uh if that scored much lower than than u or much worse off so to speak has much higher expected free energy than than another policy which says well don't do that you know it's at least you get to stay alive and conscious and all of that. uh you know that that the the pragmatic value of not doing that uh far out seek uh exceeds the information that you might gain from uh from doing this silly thing um then ultimately you don't do the silly like you right so so that's um sort of a crude way of saying that it's like the so given that this is scored over all policies which is another significant issue and why we use variation like the significance of using variational methods not just for bi biological plausibility but you know imagine that there a plethora of something that of actions or policies that an agent could take right and every single one of them is getting scored so that there could be a ton of possibilities that are getting scored all simultaneously right and so for every single one of them there is an information gain term and uh and a pragmatic value term so at one point in the the textbook and and some people have interpreted this differently and so uh Daniel and I will frequently like make a comment on this. At some point in the textbook they claim that that uh with active inference and kind of

planning is inference and their framework. They say that uh sort of exploration and exploitation quote unquote naturally balance uh one another. And that's that's not exactly right because you could still have a lot of information gained but zero pragmatic value, right? Um as in the silly example I gave a minute ago of someone knocking themselves out or something. Um you could have a full a very pragmatic very high pragmatic value and no information gain which someone who follows a routine can like kind of intuit it. It's like, oh, this is a very helpful routine and I'm not diverging from it, so I'm never really learning anything new, but uh it's because it works for me, right? Or maybe you arrived at that policy and kind of chiseled it out over time. It's like this is what my ideal day would look like, or this is what the ideal series of steps or actions I would take over the course of day would look like. Maybe with some various degrees of divergence, maybe with various degrees of like, you know, some something could come up or, you know, an unexpected events, a call on the phone, a meeting was canceled or otherwise I could get sick. But, um, you know, any of that. But the idea is sort of planning is inference here is just sort of over time being able to sort of discretise time into a sequence of steps and thinking of uh for each of these potential sequences of steps what is there to be gained and what is there to be learned most sort of intuitive sense that's how I would say they do. So a lot of these these kinds of concepts that we're introduced to on so on page 37 they figure 2.6 six they tried to essentially show again with the sort of interdisciplinary aspect of active inference uh its relation to to other fields and so we there are aspects of like control theory here in cybernetics whenever there's economics is implicated whenever we're talking about risk um so I wasn't going to break any of this down too explicitly but in part because they already kind of do that themselves but I was just going refer to this for anyone who's interested in tying this into sort of other fields and uh yeah sorry I mean I noticed your hand was up >> yeah thank you thank you for the great explanations uh I I wanted to track back a little bit and maybe have I have a question about the definition that they that they introduce at the end of chapter 3 uh which is related to expected uh expected free energy and they say that uh a generative model when endowed temporal depth opens the door to the minimization of expected free energy which uh entails at least a primitive form of agency. So uh as I understand it like the capacity to model alternative futures and to select among them to minimize expected free energy is uh basically what agency is what a how agency is defined is is there like more to it? uh is is like just the planning and the the the the vision of the f the the the modeling of the future. Is it the the only prerequisites for for defining agency in this framework or are there more or or do they go into more details into into how agency should be

defined should be uh should be conceptualized and uh maybe this is more of a philosophical question that is maybe out of scope for this uh discussion group. uh how does this relate to free will? uh does uh does it is is the capacity for planning and uh and prospection is it a sufficient basis for what we understand as free will is is there something more that is required but this is not you don't have to answer the second question I really the more important question I have is is really how agencies defined and uh and uh and if if they go into more details on this because it's what interests me uh right now I want to really get the sense of how agency can be defined and how it can be useful for us to to model like for example AI systems. >> All all really good questions and the way that you kind of um kind of uh kind of laid all of that out in that sequential order everything made sense. Um so I'm first it is I do think it's a a bit of a philosophical topic. Um that is not to say that those who engage in active inference should not engage in philosophy. It's a major topic and there are many people who are interested in active inference who want to discuss notions of agency, free will, uh consciousness, what does it mean to like plan or deliberate uh and things like that. I think that it's been sort of conceptualized in various ways uh even within active inference. I think that people are kind of publishing on this uh in different ways. Uh but many people have kind of commented on it. Um, I might be able to find a couple things I could share in the the chat, but there are those like uh Enasi Bolito who uh you know or or um um I think even Chris Fields who's actually someone who doesn't Enesso is a a philosopher uh first and foremost who who's written broadly on active inference and kind of discusses some of these things and there are a lot of authors uh sort of in that maloo who will also use it to comment upon uh AI and conceiving of AI and what would it mean to to have agency. I would say um this this uh this segment that you pointed out here in the in the textbook which is on uh page 59 for for um anyone who's interested in this uh which we also remember that's in chapter 3 which is the high road which is kind of starting from sort of what it means well let's not say means but what what it is to be an organism and to continue to exist uh like adaptively survive, right? So, it would I think that's part of why they do this like high versus low road. The low road being like a very almost strictly mathematical basian framework from the whenever you first start and then it's just from there that you can start viewing that as sort of like how we could model cognitive functions including perception and action and decision making versus uh the the other direction which is sort of like how do organisms persist. Um and so here we're looking at this from a very kind of organismic uh perspective. And then we get this comment on um having a you know potentially a minimal form of uh or primitive form of of agency. Um, and I

think just from what I can say is that because I while I'm very interested in philosophy, I tend to do a lot of kind of computational modeling uh and and and focus sort of more on on on that and relating it to sort of mechanistic processes. Um, from from sort of a hardline perspective that would say like no, there is no agency. Um I'm not so willing so freely able to say that. I don't I don't feel comfortable even though like that's not the way I should be saying that like that everything's intuition now. But um but yeah, I mean it's if if you were able to say like well there are particular kinds of processes involved here they tend to play out you know aside from some stochastic you know error that happens here or there noisy observations it's otherwise generally you know deterministic or semi-deterministic the idea of agency does sort of sort to kind of fall out of view. Um that said though I mean the the authors here themselves are saying that there is a form of agency I so another way of framing it could be like well if you looked at a model that had no capacity for planning uh and then you just stay within this active inference framework. So for example we just went over expected free energy. Uh expected free energy is necessarily like a form of kind of planning. uh that form of planning relates to certain kinds of homeostatic set points that one holds. So preferences that were related in that C matrix your attractor. So if you totally remove uh any kind of attractor from a system that means that it's not attracted towards anything in particular. Well, whenever you put that into a sort of biological environment where like ideally it would be uh you know if you remove any kind of attractor from a person then suddenly like they don't necessarily have any reason to their model can't infer that they have any reason to eat food to sleep you know to take care of themself in any way shape or form. Uh and so you know kind of in physical terms like without some kind of attractor and capacity for realizing that sort of attracting state uh in some way shape or form you you just kind of fall into entropy right you you dissipate within your your environment. Um that's uh that relates to a notion in active inference and really a lot of other computational modeling in general whether it's active inference or or not. But it's all to say like an organism doesn't necessarily have to have a set of beliefs that perfectly match its environment. It just needs to have the right equipment for navigating that environment well enough to to realize the attraction. So I I'm afraid I'm getting just a little distanced from from your your query, but it's just to say that I do think maybe the way they frame it here makes sense. The idea of if you're able to kind of predict a little bit more into the future for making different kinds of decisions, right? Because I even in like social theory or social sciences or things like that or philosophy people uh one one way of framing agency would be to say like well if if you have no actions available to you and you have no way of trying like being

allowed to choose from different available actions uh for whatever your own purposes are then you don't have agency. Um, and and for those who aren't even trying to take it from a philosophical standpoint, maybe even more of a legal standpoint like you or or or or moral standpoint, like you shouldn't remove someone's capacity, their agency, right? Everyone should be, you know, someone might make a claim that everyone should be entitled to some level of agency. That usually does refer back to the general idea of being able to have actions that are available to you and you get to freely choose. In that case, planning would be a form like having a capacity for planning and action would be directly related to an idea of having agency. Um, I feel like if I were to speak anymore though, I I would I might as well just like write an essay or something. Uh, so so that's that's kind of where I'll leave that. But um, I would definitely even just a few Google searches between active inference and agency or free will and things like that. There are many people who have been very interested in this topic. It's just not my particular like area of expertise. So yeah, >> thanks a lot. But I see that there's some interest in the chat and I'm also interested. Could you share the the name of the authors because I I didn't get the you said Polito or Chris Fields and PTO. >> Yeah. Yeah, for sure. While I'm doing that, if anyone else has any thoughts or questions they want to bring up, we can can definitely get to that, too. This may be off topic, but there is a paper that's co-author. I mean, Carl is the first author on this um active inference and intentional behavior. Um, and I don't know if that's like you said, it's uh um it's maybe not exactly the thing you're thinking about, but um it does discuss the difference between reactive, sentient, and intentional behaviors and simulations. So, I mean there is it's it's not in the textbook, which I know is the topic of this conversation, but um yeah, I just thought I'd mention that. >> Totally. And thank you for for sharing that. Um and it I just looked it up myself and the way you described it too. It very much it very much having not read it myself it very it seems to very much relate to the textbook which the textbook in a sense is just sort of like a snapshot of sort of where folks like Thomas Parr and Carl Friston were at in their research over the preceding decades like up to that point in time and uh and Daniel and I are frankly at this point looking forward to this new textbook that will be published next year which I think will just include a lot more that's uh been written since. Um but yeah, I mean with with intentional behavior and the way that you described like or mention like um and that's something I didn't bring up and maybe should have, but if we imagine an organism that doesn't have a capacity for planning, it then would become a much more sort of reactive organism and uh and we would view it as maybe not having agency or we would view it as not having intentional behavior. Uh so one one kind of

simple example of that is to think of like a thermostat like a control system. It's just it in that case a thermostat does have uh a homeostatic set point so to speak. We won't say it's homeostatic but we'll say it is a set point uh quite literally like that is its design. uh and so if the temperature that it reads in the room as its incoming observations um if that you know falls below the stat point then the agent the thermostat increases the temperature right but in that case it's reactive so it doesn't have to do with uh any kind of sort of you know I believe that if I do this then this will happen uh it's more of a strict like on a momentby-moment basis uh like purely reactive system right now. What someone could do is like program some kind of decision-making process within it to where oh uh you know over the next hour I'll heat up you know the thermostat will increase the temperature by one degree and then the following hour it'll increase it by 3° and then after that it'll reduce it by two degrees and that's a particular sequence of actions defining a policy and each one of them can get scored relative to some kind of attracting state like if it were to become more complex then it would start to kind of look like planning under active inference or even just planning as inference as a general scheme. Uh would someone want to attribute intentional behavior to that or agency? Um I think it's around there that like you know there are those who might disagree with one another uh over how that should be how that should be viewed. So it's around there that I'm like, yeah, it really depends on how you want to define uh agency or intentional behavior. But many organisms are in fact reactive, right? Like a lot of um kind of microc cellular organisms are essentially just sort of floating in their respective spaces and like immediately reactive to any kind of nutrients that start to touch them or something along those lines, right? Like not necessarily a capacity for planning involved. Um, so it's a very interesting topic uh to me to say the least. I just don't have any definitive statements on what is agency. Um, and then I'll just share I'll try to think of any others that that might come up. Um, and there's um this is sort of uh a quick way of sharing some of these things. But there there have been for for anyone who just kind of wants some some listening material. Um there was last year a nice series called active inference insights. I just shared the link to it. It's on like streaming services etc as well. But so so there are various figures um that were that were interviewed um uh by uh guy Darren who was involved with the institute previously and so he's done talks with um Carl Fristen and and um some other big figures in the area Mark SS Ian McGillicest who's a psychiatrist and others and so there there's a lot going on Alexander Rorbia who's more into robotics and reinforcement learning. So, it's there's kind of a nice diversity of voices uh from from different fields and um and some of them are certainly much more sort of on the

philosophical side. There's Vanjese who discusses uh consciousness but relates it to information theory but also ethical AI. Um so I would have a look at that. I think that's probably the the quickest thing I could share with you all in a single link. Um and then of course all of them have their own respective publications if you want to dive deeper. Um but uh so I see that uh we we really hit time here but um I hope that this was helpful to some of you who are still considering uh you know some of the the topics here and the the equations or formalisms or otherwise for those who are still you know kind of interested in the mathematics uh and and breaking that down further. There will be much more to come in the in the coming chapters. Uh even more so in the second half of the book whenever we start seeing particular uh p what they do is that they take different kinds of active inference simulations that have already been sort of coded up computationally and published. And so what they do is use those as sort of learning heuristics for understanding these different models. Um and so so that'll figure strongly in the second half of the book. Uh there's also the append appendix in the back of the book that uh this is the author's own take on trying to sort of describe some of the basics to probability theory such as what is the product rule or moving up to to sort of linear matrix operations that we see in uh discrete time models and um just you know essential calculus and taking derivatives and so on. So um there are many resources like essentially in the back of the book and then it's uh you're just a step away from from further learning those things on your own if you want to get any deeper from there. Uh and then yeah the the next session next week at this time we'll we will switch over to the other cohort but anyone who is here is welcome to join. We'll talk about chapter 7 which is where we'll um or excuse me that yeah uh we'll talk about chapter 7 and that will be about discrete time models. Um so a lot of the dynamics that we saw here will actually play out. We'll have a likelihood uh distribution. We will have state transitions and we'll have planning uh as as those models they they all have this kind of temporal dimension and a capacity for planning and decision- making and perhaps even agency. Uh so um yeah, I welcome you all to that and uh thank you all for for coming today and we'll we'll wrap it up there. >> Thanks so much. Andrew, when when are you doing chapter 3? It sounds like you're switching over. >> Yeah. Yeah. Yeah. There's uh so uh Daniel and I do this sort of alternating. Unfortunately, it's always a little confusing to ex like describe to people quickly, but it's essentially uh there's a morning and an evening session relative to I'm in the United States. So, um thinking that those time zones u the in he always does the morning, I do the evening. Um so, two weeks from now, it'll be chapter three with this cohort. So, it's sort of we we it's sort of just this uh back and forth. Um, >> so you'll be in the morning

on the next Thursday. Then you're just going to continue to flip up more or less. >> Yeah, I will always do the evening because that's when I'm available. But the essentially the other cohort is doing the second half of the book and then this cohort is doing the first half. So we have this kind of odd zigzagging between, you know, there it'll be uh next week it'll be chapter 7 for second half of the book. week after that would be chapter three, then chapter eight, then chapter four. You see this kind of back and forth like that. But the idea is that it sort of allows us to have two separate cohorts going simultaneously for a session before we take another month or two long break. Um, so it just kind of it's a bit of a speedrun, I guess, in that sense. But, um, anyway, yeah, so two weeks from now, chapter 3. All right. Thanks so much, everyone.